# Blush – Solution Approach Overview (Phase 1)

Salesforce Fresher Batch | Milestone 2: Solution Overview | 23 Oct 2026 | For mentor review

## 1. Purpose

This is a short overview of how I plan to build Phase 1 of the Blush Retail & Wholesale Management project. It covers the approach for each requirement, the design principles and best practices I will follow, and the assumptions I have taken where the assignment is unclear, with the reason for each. Detailed object and field design will follow once this direction is approved. Q-numbers refer to the Milestone 1 document (Clarifying Questions and Assumptions).

## 2. Solution at a Glance

| Req | Requirement | Approach | Built with |
|---|---|---|---|
| A | Customers and sales | Lead, Account, Contact, Opportunity and Order with Retail and Wholesale record types; automatic customer matching from the Opportunity | Declarative + Apex |
| B | Campaigns | Standard Campaign object with a “Active This Month” view for Marketing | Declarative |
| C | Products and pricing | Product catalogue, Standard, Retail and Wholesale price books, automatic price book setup for new products | Declarative + Apex |
| D | Follow-up reminders | Nightly scheduled job that creates one reminder Task per qualifying Opportunity | Apex (Batch) |
| E | Commission | Calculated per order and stored on the Order | Apex |

| Req | Requirement | Approach | Built with |
|---|---|---|---|
| F | Order controls | Order locked for Sales once Paid | Validation + Apex guard |
| G | Invoice, discount, GST | Invoice PDF from the Order, discount approval above 10%, CGST/SGST/IGST calculation | Visualforce + Approval + Apex |
| H | Reports and dashboards | Eight reports and three team dashboards | Declarative |
| I | Users and app | Two groups using permission sets, and a “Blush” Lightning app as default | Declarative |
| J | Quality and testing | Apex tests aiming for 100% coverage, plus functional, edge, error and security tests | Apex tests + test scripts |

# 3. Approach for Each Requirement

## A. Customer and Sales Management

Retail and wholesale are separated by **record types** on Account, Opportunity and Order, each with its own layout and list views, so they are always clearly distinguishable. Opportunity gets custom fields for customer name, email and phone, because Opportunity has no native fields for these.

When an Opportunity is created, an Apex trigger checks for an existing Lead by email first, then phone (phone compared on the last 10 digits). If a Lead matches, it is converted. If not, the Account (named after the customer) and Contact are created from the Opportunity, with the record type taken from the Opportunity. Existing Accounts are reused rather than duplicated. A “Create Order” action then builds the Order and copies the products across.

**Why Apex:** matching on two keys with a priority order, and Lead conversion, cannot be done with standard Flow actions, and it must work for many records at once.

## B. Promotions and Campaigns

Standard Campaign object, managed by users with the Marketing User checkbox. A formula field marks a campaign as active this month if it overlaps the current month, and a list view and report use it, so Marketing sees current-month campaigns without manual filtering.

## C. Product Catalogue and Pricing

The ten valid products from Products.xlsx are loaded into Product2 with Category, Cost Price, and two checkboxes (Available for Retail, Available for Wholesale). Base price is the retail price; wholesale price is 30% lower. Three price books are used: Standard (required by Salesforce), Retail and Wholesale.

A trigger on Product2 creates and updates price book entries automatically when a channel is ticked or the price changes, so Sales can use new products with **no admin setup**. Unticking a channel deactivates the entry instead of deleting it, so past orders stay intact. Opportunity record type, price book and Account type are kept consistent by a default and a validation rule.

## D. Follow-up Reminders

A scheduled Apex batch runs each night. For each open Opportunity it looks for incomplete Tasks or Events dated between five days ago and today. If one exists and no reminder was sent before, it creates one high-priority Task for the Opportunity Owner and marks the Opportunity so it is never reminded twice. The job ignores its own reminder Tasks and logs failures so they retry the next night.

## E. Order Commission

Commission is calculated per order, on the order amount after discount and before GST: 1% if it is Rs. 1,000 or less, otherwise a flat Rs. 30. The result is stored on the Order and frozen once the order is Paid. The thresholds sit in Custom Metadata, so a rule change needs no code change.

## F. Order Controls

Sales users can edit an order while it is being processed. Once it is Paid, validation rules block changes to the order and its products, and a small trigger blocks deletion of products. Only finance or admin users, through a custom permission, can edit a Paid order.

### G. Invoice, Discount and Tax

- **Discount:** captured as a percentage per line. Any line above 10% sends the order to the submitter's manager through a standard Approval Process, and the order cannot be confirmed until approved.
- **GST:** calculated per line on the discounted unit price. Jewellery is 3%; Garments are 5%, or 12% above Rs. 1,000. For a Maharashtra billing address the tax is split equally into CGST and SGST; otherwise IGST applies.
- **Net price:** (Unit Price − Discount) × Quantity + Taxes, as in the assignment.
- **Invoice:** a Visualforce PDF opened from a button on the Order. It follows the supplied invoice format and shows only the tax columns that apply. Store name, address and GSTIN come from a Store record.

### H. Reports and Dashboards

Eight reports: monthly commission by rep, quarterly revenue (Indian fiscal year), monthly orders, retail vs wholesale with previous month, top 3 reps by commission, total GST payable, top 10 products by quantity, and campaigns active this month. They feed three dashboards for Retail, Wholesale and Marketing.

### I. Users and Salesforce App

Both groups get the Salesforce licence, because the cheaper Platform licence excludes Lead, Opportunity and Campaign. Access comes from permission sets rather than custom profiles. A Lightning app named "Blush" is set as the default app for employees.

### J. Quality and Testing

See Section 5.

## 4. Design Principles and Best Practices

| Area | What I will follow |
| --- | --- |
| Solution design | Use standard Salesforce features first. Add custom code only where configuration cannot meet the rule, and state the reason for each case. |

| Area | What I will follow |
|---|---|
| Configurable rules | GST slabs, commission rule, discount limit, home state and reminder window live in Custom Metadata, not in code. No hardcoded Ids or names. |
| Coding | One trigger per object with no logic inside it; logic in handler and service classes; selector classes for queries; small, documented methods with ApexDoc comments. |
| Bulk and performance | Every trigger and job handles 200 records, with no queries or DML inside loops. Jobs can be re-run safely without creating duplicates. |
| Error handling | Partial-success DML where one bad record must not block others, clear messages for users, and failures written to a log object. |
| Security | Minimum-access profile plus permission sets. Private sharing for Opportunities and Orders. Calculated fields are read-only. Cost price is hidden from Sales and Marketing. Code respects sharing and field security, with any exception documented. |
| Data integrity | Tax, totals and commission are stored on the Order, so later rule or price changes never alter past invoices. |
| Testing | Test classes written together with the code, positive, negative and bulk cases, real permission sets in tests, no reliance on existing org data. |
| Documentation and delivery | Source in Git with a branch per requirement, descriptions on every field and flow, a data dictionary, a user guide and a test report. |

## 5. Testing Approach

- **Apex unit tests** for every class, with the goal of 100% coverage and tests that check behaviour, not just execution. The platform minimum of 75% is a floor, not the target.

- **Functional tests** for normal, edge and error cases, written as scripts and recorded with screenshots. Examples: matching on email only, phone only, and different leads for each; GST at exactly Rs. 1,000 and Rs. 1,000.01; discount of exactly 10% and 10.01%; reminder activities dated 5 and 6 days ago.
- **Bulk tests** with 200 records for customer linking, price book sync and order calculation.
- **Security tests** confirming that Sales cannot edit Paid orders, Marketing cannot see Orders, and cost price stays hidden.
- **Report checks** against a small demo data set with hand-calculated totals.
- **Evidence for review:** test log, coverage report, sample invoices (intra-state and inter-state) and dashboard screenshots.

# 6. Assumptions Taken and Why

## 6.1 Assumptions that shape the data model

| Unclear point | Assumption | Why |
|---|---|---|
| Who is the Sales Representative? | The Opportunity Owner, copied into a fixed field on the Order | The owner works the deal, and a fixed copy keeps commission stable if ownership changes |
| Person Accounts or Business Accounts? | Business Account plus Contact, Account Name is the customer's full name | Person Accounts cannot be switched off and complicate Lead conversion |
| Are prices tax-inclusive? | Tax-exclusive | Matches the assignment formula; the sample sheet differs and is treated as an error |
| GST slab test for garments (Rs. 1,000)? | Applied to the discounted unit price per piece | Quantity should not push a cheap item into the higher slab |
| Commission per line or per order? | Per order | The assignment says commission is per order |

| Unclear point | Assumption | Why |
|---|---|---|
| What is “net amount received”? | Order total after discount, excluding GST, with full payment assumed | Keeps the rule simple; partial payments are a later phase |
| Is there a Store? | Yes, a custom Store object linked to the Order | The invoice needs store name, address and GSTIN |
| Order statuses | Draft = processing, Activated = confirmed, Paid = locked | Uses standard fields and keeps reporting simple |

## 6.2 Assumptions for customers, pricing and approvals

| Unclear point | Assumption | Why |
|---|---|---|
| Email and phone match different Leads | Email wins; the other Lead is flagged as a possible duplicate | Email is more unique than phone, and flagging avoids silent data loss |
| How are phone numbers compared? | Last 10 digits | Handles +91, spaces and dashes |
| Existing Account or Contact found | Reuse it | Prevents duplicate customers |
| Record type conflict between Lead and Opportunity | The Opportunity’s record type wins | The Opportunity is the sales decision for the deal |
| Product availability by channel | Two checkboxes; entries created automatically in the matching price book | Native price book rules block unavailable products |
| Discount limit | 10% per line, as a percentage | Makes the check exact and avoids rounding disputes |
| Who approves? | The submitter’s manager; the order is locked while pending | Uses the role hierarchy and prevents changes during review |
| Can Paid orders ever be edited? | Only finance or admin through a custom permission | Gives a controlled override |

## 6.3 Assumptions for reminders, reports and delivery

| Unclear point | Assumption | Why |
|---|---|---|
| “Within the last five days” | Today minus 5 days up to today, future dates ignored | Matches the 15 July example in the assignment |
| One reminder per Opportunity | Once ever, not once per run | Literal reading, simplest to build and test |
| Quarter definition | Indian fiscal year, April to March | Matches Indian business reporting |
| Revenue figures | Demo prices (cost 100, retail 200, wholesale 140) used on a demo product; revenue excludes GST | Reproduces the assignment’s figures while real orders use real prices |
| GST payable | Output GST only | Input tax credit is outside the assignment |
| Net sales quantity | Returns not deducted; cancelled orders excluded | Goods are not taken back, and cancelled orders are not real sales |
| Campaigns for this month | Any campaign active at any point in the month | A running campaign that started earlier should still show |
| Licences and org | Developer Edition, Salesforce licence for both groups | Platform licences exclude Lead, Opportunity and Campaign |
| Invoice number and PDF | Auto-number (INV-0001) and a Visualforce PDF | Native, no extra licence; gaps from abandoned drafts are acceptable |
| Rounding | Half-up to 2 decimals per line; CGST is half and SGST is the remainder | Parts always add up to the total |
| Sample data issues | Blank last product row dropped; “Jwellery” corrected to “Jewellery”; all ten products available to both channels | Needed to load clean data |

# 7. New Points for Mentor Decision

These came up while designing and were not covered earlier. I will proceed with the proposal unless advised otherwise.

1. **When is an Opportunity processed?** Proposal: when it is created, with a retry option.
2. **Which orders count for commission, revenue and GST?** Proposal: Activated and Paid only; Draft and Cancelled are excluded.
3. **An existing Account has a different record type** (for example, a retail customer placing a wholesale order). Proposal: create a new Account of the Opportunity's type.
4. **Sharing model.** Proposal: Leads, Opportunities and Orders private; Accounts readable by all Sales users.
5. **Orders per Opportunity.** Proposal: one Order per Opportunity in Phase 1.
6. **Discount entry.** The sample sheet shows discount as an amount; I will capture a percentage.
7. **Platform check.** Standard Orders may restrict edits once Activated. I will verify this first, and if it blocks normal sales work I will use a separate payment status field.

On approval, implementation starts with org setup, data model, security and the product and price book setup.